

Complementarity in Social Measurement: A Partition-Logic Approach

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Partition logics—non-Boolean event structures obtained by pasting Boolean algebras—provide a natural language for situations in which a system has a definite latent state but can be accessed and resolved only through mutually incompatible coarse-grained modes of observation. We show that this structure arises in a range of social-science settings by constructing six explicit examples from personnel assessment, survey framing, clinical diagnosis, espionage coordination, legal pluralism, and organizational auditing. For each case we identify the latent state space, the observational contexts as partitions, and the shared atoms that intertwine contexts, yielding instances of the L_{12} bowtie, triangle, pentagon, and automaton partition logics. These examples make precise a notion of *social complementarity*: different modes of inquiry can be incompatible even though the underlying system remains fully value-definite. Complementarity in this sense does not entail contextuality or ontic indeterminacy. We further compare the classical probabilities generated by convex mixtures of dispersion-free states with the quantum-like Born probabilities available when the same exclusivity graph admits a faithful orthogonal representation. The framework thus separates logical structure from probabilistic realization and suggests empirically testable benchmarks for quantum-cognition models.

I. INTRODUCTION

Complementarity, the impossibility of simultaneously observing all relevant properties of a system, is usually associated with quantum mechanics. Niels Bohr famously argued that the wave and particle aspects of a quantum object cannot be revealed in a single experimental arrangement. Yet the *formal* structure underlying complementarity—a collection of Boolean “classical mini-universes” pasted together at shared elements—appears naturally in many social-science measurement situations. In these situations the system under study (an applicant, a survey respondent, an organization) possesses fully determinate properties, but the observer is forced to choose among incompatible modes of inquiry, each of which reveals only a coarse-grained view of the true state.

The mathematical framework that captures this situation is the *partition logic* [1–4]. A partition logic is built from a finite set of underlying types $S_n = \{1, 2, \dots, n\}$. Each “context” or “block” is a partition of S_n into groups; elements in the same group are those types that *cannot be distinguished* when that particular observational context is chosen. Two contexts are then “pasted” together whenever they share an identical group of types—an “intertwining atom.” The resulting algebraic structure is, in general, non-Boolean: the join or meet of propositions from different contexts need not be defined, reflecting the fact that one cannot combine information from incompatible observations in a straightforward logical way.

The purpose of this paper is to make this abstract formalism concrete through six social-science scenarios that

span organizational psychology, political science, clinical psychology, security studies, comparative law, and public administration. For each scenario we will answer three questions explicitly:

1. What are the **types** (elements of S_n), and what do they represent in the social world?
2. What are the **contexts** (partitions), and what observational procedure does each one correspond to?
3. What are the **intertwining atoms** (shared partition elements), and why does the same group of types happen to be indistinguishable under two different procedures?

We will also discuss a remarkable mathematical fact: on every such partition logic, *two* distinct probability theories can be defined. The first is the “natural” classical one, in which probabilities are convex combinations of dispersion-free (0/1) truth assignments. The second is a “quantum-like” Born-rule probability computed from a faithful orthogonal representation of the graph. These two probability theories make different numerical predictions, and the difference is in principle empirically testable—a fact of considerable interest for the quantum-cognition program [5, 6]. There may exist more than two probability theories satisfying Kolmogorov-type axioms on blocks (i.e., classical mini-universes); in particular, (i) exclusivity (probabilities of mutually exclusive events sum up), (ii) completeness (probabilities of mutually exclusive events sum up to 1). For an explicit example, we shall encounter a third type when discussing pentagon/pentagram logics.

Section II reviews the formalism. Sections III–VIII present the six scenarios. Section IX discusses the dual probability interpretation. Section X concludes.

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II. PRELIMINARIES: PARTITION LOGICS AND THEIR PROBABILITIES

A. The elements: latent states, observed categories, and pasting

A partition logic starts from a finite set $S = \{s_1, s_2, \dots, s_n\}$ of *latent states*. Each latent state is one fully specified possibility in the model: for example, a voter who is economically left-leaning and culturally conservative, or a household that is above the poverty line but materially deprived, or a job applicant whose true competence profile is “strong analytical skills, weak interpersonal skills.” We assume that, in reality, the system under study is exactly one latent state $s \in S$, even if the observer cannot always determine which one. The substantive interpretation of a latent state—what it means for a person, household, student, or organization to be in state s —is called the *latent social profile* associated with s . The word “latent” signals only that the full profile is not directly read off from any single observational procedure; it does *not* imply that the profile is indeterminate or unreal.

Potentially, the set S of latent states captures the *ontology* of the situation: what the system truly is. By contrast, what can be *inferred* about the system—and what distinctions cannot be drawn with a particular instrument—belongs to the *epistemology*. That epistemological structure is encoded by partitions of S .

A *context* is a partition $\mathcal{C} = \{B_1, B_2, \dots, B_m\}$ of S into m nonempty, pairwise disjoint subsets whose union is S . Each context represents one available mode of observation: a survey item, an administrative classification rule, a diagnostic instrument, a coding scheme, or a framing device. Each cell $B_j \subseteq S$ of the partition is called an *observed category* (synonymously, an *atom* in the language of partition logic). An observed category is the coarsest unit of information that the chosen context can deliver: when the observer selects context $\mathcal{C}$, all latent states within the same observed category B_j produce indistinguishable outcomes. The observer learns only *which observed category* the true latent state belongs to, not which specific latent state it is. Thus the partition encodes the resolving power—and the limitations—of a particular mode of inquiry.

To summarize the ontology–epistemology mapping:

- A **latent state** $s \in S$ is one exact underlying possibility (ontology).
- A **latent social profile** is the substantive social-science interpretation of that latent state—what it means to be a Star applicant, a Cognitive-dominant patient, a financially sound organization, etc. (ontology, interpreted).
- A **context** $\mathcal{C}$ is a partition of S induced by a particular instrument, survey frame, legal code, or audit directive (epistemology: choice of observational

mode).

- An **observed category** $B_j \subseteq S$ is one cell of that partition—the group of latent states that the chosen context cannot tell apart (epistemology: what is actually seen).

Two contexts $\mathcal{C}$ and $\mathcal{C}'$ may share an observed category: a subset $B \subseteq S$ may appear as an atom in both partitions. (They may, in principle, share more than just one atom corresponding to more than just one latent state.) This means that there is a group of latent states that “look the same” regardless of which of the two observational procedures is used. Such a shared observed category is called an *intertwining atom*. It is the formal point at which the two Boolean subalgebras—each representing the propositional logic of one context, in which all distinctions are classical and simultaneously decidable—are “pasted” together to form the full, non-Boolean partition logic [7–9].

Complementarity manifests itself in the fact that different contexts partition the latent-state space differently: observed categories that are resolved (split into finer groups) by one context may be conflated (merged into a single group) by another. Crucially, every latent state retains its determinate identity throughout—the complementarity is a limitation on the *observer’s* ability to ascertain the latent social profile, not an indeterminacy in the profile itself.

B. Graphical representations

Two equivalent graphical notations are common. In a (uniform) hypergraph [10] (often also referred to as a *Greechie orthogonality diagram*), each context is drawn as a smooth line and each atom as a vertex on that line; intertwining atoms sit at the intersection of two (or more) lines. In the *exclusivity graph*, vertices are atoms and edges connect atoms that are mutually exclusive (i.e., belong to the same context).

C. Multiple probability theories on the same structure

Henceforth, any admissible probability measure must satisfy two Kolmogorov-type conditions on each block (i.e., classical mini-universe) [11]:

1. **Additivity/exclusivity:** the probability of a union of mutually exclusive events within the same block is the sum of their individual probabilities (cf. Cauchy’s functional equation [12]); and
2. **Normalization/completeness:** the probabilities of all mutually exclusive events comprising a block must sum to 1.

A *dispersion-free state* (or two-valued weight) is a function v : atoms $\rightarrow \{0, 1\}$ that assigns 1 to exactly one atom per context. Intuitively, it encodes the statement, “the system *is* this type and no other.” Every dispersion-free state corresponds to one of the n latent types in S_n .

a. Classical probabilities. Given a population of systems (applicants, respondents, etc.), the fraction of each type defines a probability distribution over S_n . The resulting probabilities on atoms are convex combinations of the dispersion-free weights: $P(B_j) = \sum_{i \in B_j} \lambda_i$, where $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$. The set of all such distributions forms the *vertex packing polytope* $VP(G)$ of the exclusivity graph [13]—a convex polytope whose vertices are the dispersion-free states.

b. Quantum-like (Born-rule) probabilities. If the exclusivity graph G admits a *faithful orthogonal representation* (FOR)—an assignment of unit vectors $|v_i\rangle$ in $\mathbb{R}^d$ to atoms such that vectors within the same context are orthogonal and vectors from different contexts are non-orthogonal [14, 15]—one can define a “probability” by choosing a unit “state” vector $|c\rangle$ and setting

$$P_{\text{Born}}(c, v_i) = |\langle c | v_i \rangle|^2. \quad (1)$$

By the Pythagorean theorem, within each context the probabilities add to one. The set of all distributions obtainable this way is the *theta body* $TH(G)$ [13], which in general *strictly contains* $VP(G)$: there exist Born-rule distributions that cannot be realized by any classical mixture of dispersion-free weights.

Crucially, the same formal graph can be implemented by a social-science “black box” (yielding classical probabilities from $VP(G)$) or by a quantum resource (yielding Born-rule probabilities from $TH(G)$). The graph alone does not determine the probability theory; the *resource inside the box* does [9].

See Section V E for a discussion of another “exotic” type of probability measure, as exposed by Gerelle, Greechie, and Miller [7, Fig. V] as well as Wright [8].

III. SCENARIO 1: COMPLEMENTARY PERSONNEL ASSESSMENTS (L_{12})

In what follows, we develop several concrete instances and enactments of the formal scenario involving the *partition logics* introduced earlier.

For related discussions and additional examples, see the works of Khrennikov [16–20] and other researchers [21–24], who have explored similar quantum-like structures in cognition and decision-making.

A. What are the types?

A firm’s human resources (HR) department faces a pool of job applicants. Each applicant has a true, latent *competence profile*, falling into one of four categories:

- **Type 1 (“Star”)**: Excellent both in analytical reasoning and in interpersonal communication.
- **Type 2 (“Analyst”)**: Strong analytical skills but weaker interpersonal skills.
- **Type 3 (“Communicator”)**: Strong interpersonal skills but weaker analytical ability.
- **Type 4 (“Developing”)**: Still developing in both dimensions.

Thus $S_4 = \{1, 2, 3, 4\}$, and every applicant *is* exactly one type—the uncertainty is the HR department’s, not the applicant’s.

B. What are the contexts?

Two assessment instruments are available:

- **Written aptitude test (C_W)**: A timed analytical exam. It can identify Stars (type 1) by their top scores and Analysts (type 2) by their strong-but-not-top scores, but it cannot distinguish Communicators from Developing applicants, because both score poorly on a purely analytical task. The three observable outcomes are therefore: “Outstanding” = $\{1\}$, “Adequate” = $\{2\}$, “At-risk” = $\{3, 4\}$.
- **Behavioral interview (C_I)**: A structured interpersonal exercise. It can identify Stars (type 1) by their social poise, and Communicators (type 3) by their good-but-not-top social skills, but it cannot distinguish Analysts from Developing applicants, because both perform weakly in interpersonal tasks. The three outcomes are: “Outstanding” = $\{1\}$, “Adequate” = $\{3\}$, “At-risk” = $\{2, 4\}$.

Crucially, suppose the two instruments **cannot be administered independently** to the same applicant. Administering the written test first induces “stereotype threat” or test anxiety that contaminates subsequent interview performance; conversely, the social priming of an interview biases subsequent test-taking behavior (anchoring effects [25]). The HR department must therefore *choose one instrument per applicant*—this is the source of complementarity.

TABLE I. Response profiles for four applicant types under two complementary instruments. Each row is an applicant type; each column shows how that type appears under the given instrument.

Applicant type	Written test outcome	Interview outcome
1: “Star”	Outstanding	Outstanding
2: “Analyst”	Adequate	At-risk
3: “Communicator”	At-risk	Adequate
4: “Developing”	At-risk	At-risk

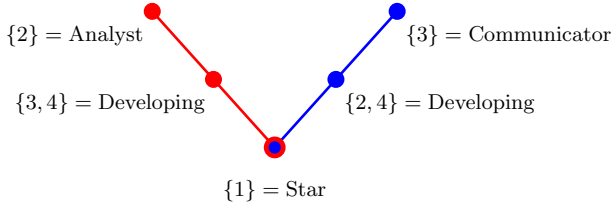


FIG. 1. Greechie orthogonality diagram (hypergraph) of the L_{12} personnel-assessment logic. Each line represents one context—one available mode of observation. The **written-test context** C_W (left) carries three observed categories: the singleton $\{2\}$ (Analyst, recognizable by adequate test performance), the pair $\{3, 4\}$ (Communicator and Developing, conflated because both score poorly on the analytical task), and the singleton $\{1\}$ (Star, top scorer). The **interview context** C_I (right) carries a different triple of observed categories: $\{3\}$ is now resolved (Communicator, identifiable by social poise) but $\{2, 4\}$ is conflated (Analyst and Developing, both weak in interpersonal tasks). The observed category $\{1\}$ is the *intertwining atom*: it appears in both contexts because the Star’s latent social profile—excellence on both dimensions—makes this profile identifiable regardless of which instrument is chosen. The five observed categories, together with their lattice-theoretic joins and meets, generate a 12-element non-Boolean logic, hence the name L_{12} .

Formally, the two contexts are represented by the partitions $\{\{1\}, \{2\}, \{3, 4\}\}$ and $\{\{1\}, \{3\}, \{2, 4\}\}$, or, in epistemic terms,

$$\begin{aligned} &\{\text{Star, Analyst, Developing}\} \\ &\text{and} \\ &\{\text{Star, Communicator, Developing}\}. \end{aligned} \quad (2)$$

C. What is the intertwining atom?

The atom $\{1\}$ (“Outstanding”) appears identically in both partitions (Table I). This is because the Star (type 1) excels on *both* dimensions, so regardless of which instrument is chosen, Stars are singled out. In the language of partition logic, $\{1\}$ is the intertwining atom at which the two three-element Boolean algebras 2^3 are pasted together.

Note what is *not* shared: the “At-risk” label has a *different meaning* in each context— $\{3, 4\}$ under the written test vs. $\{2, 4\}$ under the interview. Although the label is the same, the underlying sets of types are different, so these are *not* intertwining atoms. Only atoms that correspond to exactly the same set of types in both contexts are identified.

Formally, the two contexts, depicted in Fig. 1, are thus represented by the partitions $\{\{1\}, \{2\}, \{3, 4\}\}$ and $\{\{1\}, \{3\}, \{2, 4\}\}$, or, in epistemic terms,

$$\begin{aligned} &\{\text{Star, Analyst, Developing}\} \\ &\text{and} \\ &\{\text{Star, Communicator, Developing}\}. \end{aligned} \quad (3)$$

D. What does the partition logic tell us?

The partition logic L_{12} (Fig. 1) encodes a precise trade-off. By choosing the written test, the HR department gains the ability to distinguish type 2 from types 3 and 4, at the cost of being unable to distinguish type 3 from type 4. By choosing the interview, it can distinguish type 3 from types 2 and 4, but now type 2 and type 4 are conflated. No single instrument resolves all four types. This is *complementarity*: the information about one aspect of the applicant (analytical vs. interpersonal) comes at the expense of information about the other. Yet every applicant *has* a determinate type; the partition logic captures an epistemic limitation of the observer, not an ontic indeterminacy of the system.

IV. SCENARIO 2: FRAMING EFFECTS IN PUBLIC OPINION SURVEYS (PENTAGON LOGIC)

A. What are the types?

A political psychologist studies a population whose members differ in their latent configuration of attitudes across five policy domains. Rather than continuous attitude scales, suppose (for simplicity) that the relevant attitudinal differences can be captured by $n = 11$ discrete respondent *profiles*, denoted $S_{11} = \{1, 2, \dots, 11\}$. Each profile specifies a particular pattern of beliefs about immigration, healthcare, taxation, education, and defense spending. For example:

- Profile 1 might be a “consistent moderate” who holds centrist views on all five issues.
- Profile 7 might be a “libertarian” who favors minimal government across all domains.
- Profile 10 might be a “hawk-dove” who supports high defense spending as well as generous social programs.

Suppose the 11 profiles are exhaustive and mutually exclusive: every respondent *is* exactly one profile.

B. What are the contexts?

The five policy issues define five *survey contexts*. In each context, the respondent is asked a block of questions about one issue (e.g., immigration), and their response is classified into one of three *opinion clusters*: (A) supportive, (B) moderate, (C) opposed (or similar trichotomy). Crucially, due to well-documented **question-order and framing effects** [26], the way respondents categorize their own position on an issue is influenced by the issue discussed immediately before. In survey-methodology terms, asking about immigration “primes”

concepts (e.g., national identity, fiscal cost) that alter responses to a subsequent healthcare question.

For this reason, in any single session a respondent can be meaningfully polled on only *one* issue—attempting to cover all five in sequence would yield contaminated data that does not reflect the respondent’s true underlying profile. This forced choice of survey frame is the source of complementarity.

The five contexts, adapted from the partition logic of the pentagon [7–9], are:

$$\begin{aligned}
 C_1 \text{ (immigration)} &= \underbrace{\{1, 2, 3\}}_{\text{Cluster A}}, \underbrace{\{4, 5, 7, 9, 11\}}_{\text{Cluster B}}, \underbrace{\{6, 8, 10\}}_{\text{Cluster C}}, \\
 C_2 \text{ (healthcare)} &= \{6, 8, 10\}, \{1, 2, 4, 7, 11\}, \{3, 5, 9\}, \\
 C_3 \text{ (taxation)} &= \{3, 5, 9\}, \{1, 4, 6, 10, 11\}, \{2, 7, 8\}, \\
 C_4 \text{ (education)} &= \{2, 7, 8\}, \{1, 3, 9, 10, 11\}, \{4, 5, 6\}, \\
 C_5 \text{ (defense)} &= \{4, 5, 6\}, \{7, 8, 9, 10, 11\}, \{1, 2, 3\}.
 \end{aligned} \tag{4}$$

Each context partitions all 11 profiles into three clusters.

C. What are the intertwining atoms?

Consider contexts C_1 (immigration) and C_2 (healthcare). The atom $\{6, 8, 10\}$ appears as Cluster C in C_1 and as Cluster A in C_2 . Substantively, this means that profiles 6, 8, and 10 are respondents who (i) oppose the prevailing immigration policy *and* (ii) support the prevailing healthcare policy—and *both of these facts are observable regardless of whether the survey asks about immigration or healthcare*. The cluster $\{6, 8, 10\}$ is an intertwining atom precisely because these three profiles happen to “look the same” under both frames.

The five intertwining atoms, one per adjacent pair of contexts, form the outer vertices of the pentagon diagram (Fig. 2). They are: $\{1, 2, 3\}$ (shared by C_1 and C_5), $\{6, 8, 10\}$ (shared by C_1 and C_2), $\{3, 5, 9\}$ (shared by C_2 and C_3), $\{2, 7, 8\}$ (shared by C_3 and C_4), $\{4, 5, 6\}$ (shared by C_4 and C_5).

Formally, the five cyclically intertwined contexts (or cliques) form a pentagon/pentagram logic [8, p. 267, Fig. 2] supporting 11 dispersion-free states. Constructing the five contexts from the occurrences of the dispersion-free value 1 on the respective ten atoms results in the partition logic:

$$\begin{aligned}
 &\{\{1, 2, 3\}, \{4, 5, 7, 9, 11\}, \{6, 8, 10\}\}, \\
 &\{\{6, 8, 10\}, \{1, 2, 4, 7, 11\}, \{3, 5, 9\}\}, \\
 &\{\{3, 5, 9\}, \{1, 4, 6, 10, 11\}, \{2, 7, 8\}\}, \\
 &\{\{2, 7, 8\}, \{1, 3, 9, 10, 11\}, \{4, 5, 6\}\}, \\
 &\{\{4, 5, 6\}, \{7, 8, 9, 10, 11\}, \{1, 2, 3\}\},
 \end{aligned} \tag{5}$$

as depicted in Fig. 2.

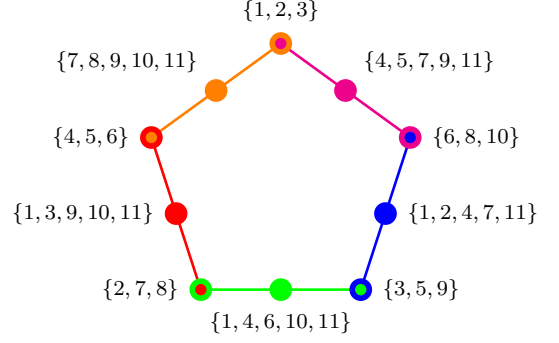


FIG. 2. Pentagon survey logic. Each color corresponds to a context (policy issue) corresponding to stances on immigration, healthcare, taxation, education, and defense spending (clockwise from top right). The cyclic structure reflects the pattern of framing overlaps among the five issues.

D. Why the pentagon structure matters: classical vs. quantum-like bounds

Each intertwining atom is a subset of the 11 profiles. The *classical* probability that a randomly drawn respondent falls in any one of these subsets is simply the fraction of the population belonging to those profiles. The key constraint is that the five intertwining-atom subsets overlap: for instance, profile 3 belongs to both $\{1, 2, 3\}$ and $\{3, 5, 9\}$. Classical counting shows that each of the 11 profiles can contribute to at most two of the five intertwining atoms, so

$$\sum_{j=1}^5 P(\text{intertwining atom}_j) \leq 2. \tag{6}$$

This is the Klyachko–Bub–Stairs inequality [27, 28]. It says: no matter how the population is distributed across the 11 profiles, the total “intertwining-atom weight” cannot exceed 2.

If, however, the respondents’ cognitive states are modeled as vectors in a Hilbert space—as proposed in the quantum-cognition literature [5]—then probabilities are $|\langle c|v_j\rangle|^2$ and can be tuned (by choosing the state $|c\rangle$) to yield

$$\sum_{j=1}^5 P(\text{intertwining atom}_j) \leq \sqrt{5} \approx 2.236. \tag{7}$$

This 12% increase over the classical bound is an empirically testable prediction. A survey experiment that (i) implements the pentagon exclusivity structure and (ii) finds aggregate intertwining-atom probabilities exceeding 2 would constitute evidence against a classical partition-logic model and in favor of a quantum-like cognitive model.

V. SCENARIO 3: DIAGNOSTIC COMPLEMENTARITY IN CLINICAL PSYCHOLOGY (TRIANGLE LOGIC)

A. What are the latent states?

A clinical psychologist suspects that patients presenting with generalized anxiety disorder (GAD) actually fall into four latent states—four diagnostic subtypes that differ along two dimensions (cognitive worry and physiological arousal) but are never fully revealed by any single instrument:

- **Latent state 1 (“Cognitive-somatic”):** The patient experiences both high cognitive worry (rumination, catastrophizing) *and* high physiological arousal (elevated cortisol, reduced heart-rate variability). This is the most floridly symptomatic subtype.
- **Latent state 2 (“Cognitive-dominant”):** High cognitive worry but low physiological arousal. The patient ruminates intensely but does not show the somatic signature of anxiety.
- **Latent state 3 (“Somatic-dominant”):** Low cognitive worry but high physiological arousal. The patient reports feeling “fine” yet shows marked autonomic dysregulation.
- **Latent state 4 (“Subclinical”):** Moderate levels of both worry and arousal that fall below typical clinical thresholds on *either* dimension taken alone. This subtype is the hardest to classify, because it never produces a salient signal on any single instrument.

The set of latent states is thus $S = \{1, 2, 3, 4\}$. Each patient *is* exactly one subtype; the diagnostic challenge is that no single instrument can identify all four.

B. What are the three contexts?

Three diagnostic instruments are available, each resolving two of the three “clear” subtypes (1, 2, 3) as singleton observed categories while *conflating* the third with the elusive 4:

- **Context $\mathcal{C}_A$ — Self-report questionnaire (cognitive focus).** The patient completes a standardized inventory of worry-related cognitions. Instrument A picks up the cognitive dimension:
 - 1 (Cognitive-somatic) reports extreme worry $\rightarrow$ observed category $\{1\}$ (“severe cognitive”).
 - 2 (Cognitive-dominant) reports high worry $\rightarrow$ observed category $\{2\}$ (“moderate cognitive”).

- 3 and 4 both report little cognitive worry $\rightarrow$ conflated observed category $\{3, 4\}$ (“low cognitive”).

Partition: $\mathcal{C}_A = \{\{1\}, \{2\}, \{3, 4\}\}$.

- **Context $\mathcal{C}_B$ — Behavioral observation (interpersonal focus).** A clinician trained in standardized behavioral coding observes the patient during a structured social-interaction task (e.g., a simulated conversation) and systematically rates observable signs such as gaze aversion, fidgeting, tremor, and speech latency. Instrument B picks up the *behavioral expression* of anxiety:
 - 2 (Cognitive-dominant) displays visible avoidance, gaze aversion, and social withdrawal driven by ruminative worry $\rightarrow$ observed category $\{2\}$ (“avoidant”).
 - 3 (Somatic-dominant) displays visible tremor, fidgeting, and perspiration from physiological arousal $\rightarrow$ observed category $\{3\}$ (“tremorous”).
 - 1 and 4 are conflated: 1 (Cognitive-somatic) has learned to compensate socially despite high symptom load, and 4 (Subclinical) simply does not display salient signs $\rightarrow$ conflated observed category $\{1, 4\}$ (“unremarkable presentation”).

Partition: $\mathcal{C}_B = \{\{2\}, \{3\}, \{1, 4\}\}$.

- **Context $\mathcal{C}_C$ — Physiological stress test (somatic focus).** The patient undergoes a standardized stress-induction protocol while heart-rate variability and cortisol are recorded. Instrument C picks up the somatic dimension:
 - 1 (Cognitive-somatic) shows extreme autonomic reactivity $\rightarrow$ observed category $\{1\}$ (“severe physiological”).
 - 3 (Somatic-dominant) shows elevated reactivity $\rightarrow$ observed category $\{3\}$ (“moderate physiological”).
 - 2 and 4 both show unremarkable physiological profiles $\rightarrow$ conflated observed category $\{2, 4\}$ (“low physiological”).

Partition: $\mathcal{C}_C = \{\{1\}, \{3\}, \{2, 4\}\}$.

Table II summarizes the mapping. The crucial pattern is that 4 (Subclinical) *never* forms a singleton: under every instrument it is conflated with a different partner—with 3 under A, with 1 under B, and with 2 under C. It is the “stealth” subtype that each instrument misses in a different way.

C. Why only one instrument per session?

The three instruments interfere with one another in a cyclic pattern that prevents sequential administration within a single diagnostic session:

1. *A degrades B*. Completing the self-report questionnaire induces introspective self-focus, which alters the patient’s spontaneous behavior in the subsequent social interaction [29]. The behavioral observation would then reflect questionnaire-primed comportment rather than the patient’s natural presentation.
2. *B degrades C*. The social interaction of being observed by a rater elevates the patient’s baseline physiological arousal (social-evaluative threat [30]), contaminating the subsequent stress-test readings.
3. *C degrades A*. Undergoing a physiological stress-induction protocol triggers mood-congruent recall and somatic focusing [31], biasing the patient’s subsequent self-reports toward overendorsement of cognitive worry.

These interferences are not merely practical inconveniences; they are systematic disturbances that change the patient’s momentary state in ways relevant to the other instruments. The clinician must therefore *choose exactly one instrument per session*—this forced choice is the source of complementarity.

D. What are the intertwining atoms?

Each pair of contexts shares exactly one observed category (Table III):

- $\{1\}$ is an intertwining atom of $\mathcal{C}_A$ and $\mathcal{C}_C$: both the self-report questionnaire and the physiological stress test single out the Cognitive-somatic subtype, because 1’s extreme scores on both the cognitive *and* the somatic dimension make it identifiable regardless of whether one probes cognition or physiology.

TABLE II. How the four latent diagnostic subtypes map to observed categories under each of the three instruments. Each row is a latent state; each column shows the observed category it falls into under the given context. Boldface marks singleton observed categories (the instrument *resolves* that subtype); plain text marks the conflated pair (the instrument *cannot* distinguish the two subtypes in that group).

Latent state	$\mathcal{C}_A$ (Self-rep.)	$\mathcal{C}_B$ (Behav.)	$\mathcal{C}_C$ (Physiol.)
1 (Cog-som)	severe cog.	unremarkable	severe phys.
2 (Cog-dom)	moderate cog.	avoidant	low phys.
3 (Som-dom)	low cog.	tremorous	moderate phys.
4 (Subclin.)	low cog.	unremarkable	low phys.

TABLE III. Intertwining atoms of the triangle logic. Each row names a pair of contexts, the shared observed category, and the clinical reason why the subtype is identifiable under both instruments.

Context pair	Shared atom	Reason for identifiability
$\mathcal{C}_A\text{--}\mathcal{C}_C$	$\{1\}$	Extreme on <i>both</i> worry and arousal
$\mathcal{C}_A\text{--}\mathcal{C}_B$	$\{2\}$	High worry shows in report <i>and</i> behavior
$\mathcal{C}_B\text{--}\mathcal{C}_C$	$\{3\}$	High arousal shows in behavior <i>and</i> physiology

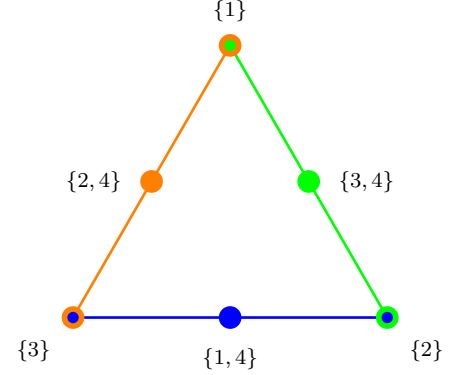


FIG. 3. Greechie diagram of the triangle logic applied to clinical diagnosis. Each edge represents one context (diagnostic instrument). The three yellow vertices are the intertwining atoms—the singleton observed categories, each identifiable under two adjacent instruments. The three white vertices are the conflated observed categories, each merging 4 (Subclinical) with a different partner. 4 is the “stealth” subtype: it never appears as a singleton under any instrument, yet it has a fully determinate latent social profile. The six observed categories, together with their lattice-theoretic operations, form a non-Boolean logic that cannot be embedded in any single Boolean algebra.

- $\{2\}$ is an intertwining atom of $\mathcal{C}_A$ and $\mathcal{C}_B$: the questionnaire flags 2 as “moderate cognitive” and the behavioral observation flags 2 as “avoidant”—different labels, but the *same set* of latent states, $\{2\}$. The Cognitive-dominant subtype is identifiable whether one reads the patient’s self-report or watches the patient’s behavior.
- $\{3\}$ is an intertwining atom of $\mathcal{C}_B$ and $\mathcal{C}_C$: the behavioral observation sees tremor and the stress test sees elevated cortisol—both pointing uniquely to the Somatic-dominant subtype.

These three intertwining atoms are the vertices of a triangle in the Greechie diagram (Fig. 3). Each edge of the triangle represents one context; the non-shared observed categories (the conflated pairs $\{3,4\}$, $\{1,4\}$, $\{2,4\}$) sit at the midpoints of the edges. The result is Wright’s *triangle logic* [32, Figure 2, p. 900] (see also [3, Fig. 6, Example 8.2, pp. 414,420,421]).

E. The exotic probability weight: clinical interpretation

Wright [8] (see also Ref. [7, Fig. V]) showed that cyclic logics with an odd number n of intertwining atoms—including the triangle, pentagon, and pentagram logics—support an “exotic” probability weight in addition to their standard dispersion-free weights. This weight assigns a value of $1/2$ to each intertwining vertex and lies strictly *outside* the classical convex hull, rendering this $\{0, 1/2\}$ -valued state inexpressible as a convex mixture of the $\{0, 1\}$ -valued dispersion-free states.

To see this, consider the sum of the weights assigned to the n intertwining atoms, $S(\omega) = \sum_{i=1}^n \omega(v_i)$. Because these atoms are arranged in a cycle where each adjacent pair $\{v_i, v_{i+1}\}$ (with $v_{n+1} = v_1$) belongs to a common block, any valid weight must satisfy the consistency constraint $\omega(v_i) + \omega(v_{i+1}) \leq 1$. For the exotic state ω_{ex} , the functional yields $S(\omega_{ex}) = n/2$. However, for any dispersion-free state $\omega_c \in \{0, 1\}$, the constraint forbids adjacent atoms from both being 1. On an odd cycle, the maximum number of non-adjacent vertices that can be assigned 1 is $(n-1)/2$; thus, $S(\omega_c) \leq (n-1)/2$. Since any state ω_{hull} in the classical convex hull is a mixture of these states, it is bounded by $S(\omega_{hull}) \leq (n-1)/2$. Because $n/2 > (n-1)/2$ for all odd n , the exotic state is strictly non-classical.

In support of the reality of such a scenario, Wright gave the following snippet [8, p. 272]:

“... suppose that there is a consumer who has no strong opinions one way or the other about the products, so that he is equally likely to say either yes or no to the first question, but suppose, further, that he is embarrassed about saying no [33] so that if he says no to the first question, then he always says yes to the second.”

What could this mean in the clinical context? It would represent a probability distribution in which there is exactly a 50% chance of observing the identifiable subtype (1, 2, or 3) under any given instrument—but this distribution is *inconsistent with any population mixture of the four diagnostic subtypes*. No matter how one adjusts the prevalence rates of the four subtypes, one cannot reproduce the exotic weight by classical mixing.

If a clinician’s empirical frequencies across a patient population approached this exotic weight, it would suggest that the classical partition-logic model (determinate subtypes, purely epistemic uncertainty) is inadequate. One possible interpretation, drawn from the quantum-cognition literature [5], is that patients’ diagnostic profiles are not fixed prior to measurement but are instead *constituted in part by the diagnostic act itself*—the instrument does not merely reveal a pre-existing subtype but participates in producing the categorization. Whether this interpretation is warranted or whether alternative explanations (e.g., systematic violations of the

single-instrument-per-session constraint) are more parsimonious is an empirical question that the partition-logic framework makes precise.

VI. SCENARIO 4: ESPIONAGE AND RELATIONAL ENCODING (CLASSICAL EPR ANALOGUE)

A. What are the types?

An intelligence agency trains agent pairs using a protocol that creates correlations between their cover stories. Each agent pair is prepared in one of four *coordination types*:

- **Type 1 (“Mirror”)**: Both agents memorize identical financial records and identical personal histories. (Code: 00— same digit in both positions.)
- **Type 2 (“Split-A”)**: Agent A memorizes a clean financial record but a fabricated personal history; Agent B memorizes the reverse. (Code: 01.)
- **Type 3 (“Split-B”)**: The reverse of Split-A. (Code: 10.)
- **Type 4 (“Contrast”)**: Both agents memorize fabricated financial records and fabricated personal histories, but with opposite details. (Code: 11.)

B. What are the contexts?

Upon capture, each agent faces one of two interrogation modes:

- **Financial background check (F)**: The interrogator examines only financial records. Agent’s response is the first digit of the code (0 = clean, 1 = fabricated).
- **Personal history interview (H)**: The interrogator examines only personal history. Agent’s response is the second digit of the code.

An agent subjected to one mode gives answers that are psychologically “activated” in a way that would contaminate a subsequent interrogation of a different mode (a well-known carryover effect in interrogation science). Thus only one mode can be applied per agent.

C. Relational encoding and why it is local

The key point is that the agency loads agent pairs from the subensemble $D = \{01, 10\}$ (Table IV), meaning only types 2 and 3 are used. The relational encoding guarantees that **if both agents face the same interrogation**

mode, their answers are always opposite: when Alice’s financial check yields “clean” (0), Bob’s yields “fabricated” (1), and vice versa. This perfect anticorrelation arises entirely from the shared codebook—a *classical common cause* established before separation.

The partition logic structure is as follows. For each agent, the two interrogation modes define two partitions of the type space. For Alice:

$$\begin{aligned} C_F^{(A)} &= \{\{1, 2\}, \{3, 4\}\} \quad (\text{first digit}), \\ C_H^{(A)} &= \{\{1, 3\}, \{2, 4\}\} \quad (\text{second digit}). \end{aligned} \quad (8)$$

For Bob, the same. Each agent’s logic is the pasting of two two-element Boolean algebras 2^2 —a minimal complementary structure.

D. Social science significance

This scenario models any situation in which two spatially or temporally separated actors produce correlated behaviors through a *shared preparation* rather than direct communication:

- **Coordinated deception:** As described—espionage cover stories.
- **Institutional scripts:** Employees trained in the same corporate protocol independently produce consistent behaviors when facing customers [34].
- **Shared cultural schemas:** Individuals socialized in the same culture give correlated survey responses about values and norms, even when separated by continents, because they internalized the same “codebook” during childhood [35].

In all cases, the correlations are *local* (no signaling) and *classical* (probabilities lie within $\text{VP}(G)$). The scenario demonstrates that correlation without communication is not uniquely quantum; what *is* uniquely quantum is the ability to *violate Bell inequalities*, which the shared-codebook mechanism can never do.

TABLE IV. Relational encoding of cover stories for agent pairs. Each entry is a two-digit code: first digit = financial record, second digit = personal history. *S* selects types whose codes match pairwise; *D* selects types whose codes differ pairwise.

Sample	Type 1	Type 2	Type 3	Type 4
<i>S</i> (same)	00			11
<i>D</i> (diff.)		01	10	

VII. SCENARIO 5: LEGAL PLURALISM AND OVERLAPPING JURISDICTIONS

A. What are the types?

In a legally pluralistic society—for example, a post-colonial state where statutory law, customary (indigenous) law, and religious law coexist [36, 37]—a citizen’s action can be classified differently depending on which legal framework is applied. Consider six types of action involving land use:

- a_1 : Clearing forest for subsistence farming on ancestral land.
- a_2 : Clearing forest for commercial logging on ancestral land.
- a_3 : Building a house on communally held village land.
- a_4 : Building a commercial structure on communally held village land.
- a_5 : Grazing livestock on a protected nature reserve.
- a_6 : Grazing livestock on privately owned pasture.

Each action is a determinate fact; the question is how it is *categorized* by different legal codes.

B. What are the contexts?

Three legal codes, each classifying actions into three categories (“Permitted,” “Restricted,” “Prohibited”), define three contexts:

- **Statutory (environmental) law (C_{stat}):** Focuses on environmental impact. Actions a_1, a_2 (forest clearing) are *Prohibited*; a_3, a_4 (building on non-forest land) are *Restricted* (requiring permits); a_5, a_6 (grazing) are *Permitted*. Partition: $\{\{a_1, a_2\}, \{a_3, a_4\}, \{a_5, a_6\}\}$.
- **Customary (indigenous) law (C_{cust}):** Focuses on ancestral rights and communal obligations. a_1, a_3 (subsistence use of ancestral/communal land) are *Permitted* (rights of use); a_2, a_5 (commercial exploitation of communal resources) are *Restricted* (requiring elder consent); a_4, a_6 (use of land outside the communal system) are outside customary jurisdiction, categorized as *Neutral*. Partition: $\{\{a_1, a_3\}, \{a_2, a_5\}, \{a_4, a_6\}\}$.
- **Religious law (C_{rel}):** Focuses on stewardship obligations. a_1, a_2 (forest clearing) are *Prohibited* (the forest is sacred); a_3, a_5 (non-destructive use) are *Permitted*; a_4, a_6 (commercial use) are *Discouraged*. Partition: $\{\{a_1, a_2\}, \{a_3, a_5\}, \{a_4, a_6\}\}$.

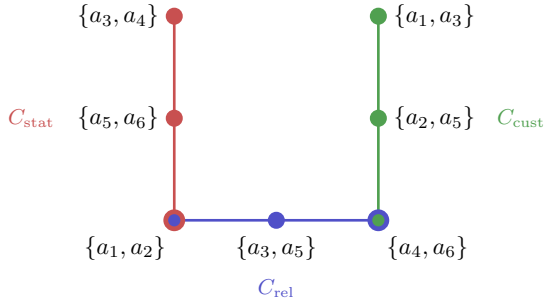


FIG. 4. Hypergraph of the legal-pluralism logic. Three legal codes (Statutory, Religious, Customary) form three contexts. Yellow vertices are intertwining atoms: $\{a_1, a_2\}$ (forest clearing—prohibited under both statutory and religious law) and $\{a_4, a_6\}$ (commercial use—set apart by both customary and religious law). White vertices represent groupings specific to one code. The non-Boolean pasting reflects the impossibility of a single “master code” that respects all three systems’ distinctions simultaneously.

C. What are the intertwining atoms?

Two atoms are shared:

- $\{a_1, a_2\}$ (“Prohibited”) is an intertwining atom of C_{stat} and C_{rel} : both statutory and religious law agree that all forms of forest clearing are prohibited, regardless of purpose. This agreement arises from a convergence of environmental and spiritual rationales.
- $\{a_4, a_6\}$ is an intertwining atom of C_{cust} and C_{rel} : both customary and religious law place commercial land use in a separate category (“Neutral”/“Discouraged”), treating it as outside the domain of traditional obligations—albeit for different reasons (customary law sees it as irrelevant; religious law sees it as morally suspect).

TABLE V. Classification of six land-use actions under three legal codes. Actions in the same cell of a given column are indistinguishable (conflated) under that code.

Action	Statutory	Customary	Religious
a_1 (subsist. clearing)	Prohibited	Permitted	Prohibited
a_2 (comm. logging)	Prohibited	Restricted	Prohibited
a_3 (house on village)	Restricted	Permitted	Permitted
a_4 (shop on village)	Restricted	Neutral	Discouraged
a_5 (graze on reserve)	Permitted	Restricted	Permitted
a_6 (graze on private)	Permitted	Neutral	Discouraged

D. What does the non-Boolean structure mean for legal practice?

In practice, a land-use dispute is adjudicated under *one* legal framework at a time—the choice of forum (statutory court, customary tribunal, or religious council) determines which distinctions are drawn and which actions are conflated:

- Under statutory law, subsistence clearing (a_1) and commercial logging (a_2) are treated identically (both prohibited), whereas customary law treats them very differently (permitted vs. restricted).
- Under customary law, a house (a_3) and a shop (a_4) on village land are in different categories, but under statutory law both are merely “restricted.”

The non-Boolean logic (Fig. 4) formalizes the well-known problem in comparative law that there is no single “master code” that simultaneously preserves all the distinctions made by each individual legal system. The choice of forum is analogous to the choice of measurement context in quantum mechanics—and it is a genuine choice with real consequences for the outcome.

VIII. SCENARIO 6: ORGANIZATIONAL AUDITING AND BOUNDED RATIONALITY (AUTOMATON PARTITION LOGIC)

A. What are the types?

A government regulatory agency must assess whether a corporation complies with regulations. The corporation can be in one of four internal *organizational states*, characterized by two binary dimensions:

- **Financial health:** Sound (0) vs. Distressed (1).
- **Operational compliance:** Compliant (0) vs. Non-compliant (1).

The four states are:

- $s_1 = (0,0)$: Financially sound *and* operationally compliant—the ideal.
- $s_2 = (0,1)$: Financially sound but operationally non-compliant—cutting corners despite having resources.
- $s_3 = (1,0)$: Financially distressed but operationally compliant—struggling but playing by the rules.
- $s_4 = (1,1)$: Financially distressed *and* non-compliant—the worst case.

Each corporation *is* exactly one of these states at any given time.

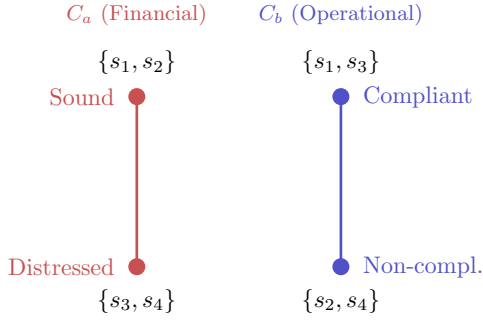


FIG. 5. The four organizational states arranged in a 2×2 grid. The financial audit (directive a , red dashed) partitions the grid horizontally: it reveals the row (financial health) but conflates columns (compliance status). The operational inspection (directive b , blue dotted) partitions vertically: it reveals the column but conflates rows. The two partitions share *no* atoms—this is maximal complementarity.

B. What are the contexts?

The auditor can probe the corporation by issuing one of two types of directive:

- **Financial audit (directive a):** The auditor requests detailed financial records. The corporation’s response reveals its *financial health* (the first digit), but the act of preparing financial disclosures causes the organization to reallocate resources internally, thereby *changing its operational compliance*. (Employees are pulled off production lines to prepare audit documents, maintenance is deferred, etc.) The auditor observes output 0 (sound) or 1 (distressed), but the original compliance state is destroyed. Partition: $C_a = \{\{s_1, s_2\}, \{s_3, s_4\}\}$.
- **Operational inspection (directive b):** The auditor sends inspectors to the factory floor. The response reveals *operational compliance* (the second digit), but the disruption of an on-site inspection causes financial perturbations (e.g., halted production, legal fees). The original financial state is destroyed. Partition: $C_b = \{\{s_1, s_3\}, \{s_2, s_4\}\}$.

C. What are the intertwining atoms?

In this scenario, the two partitions share *no* intertwining atoms (Fig. 5):

$$\begin{aligned}
 C_a &= \left\{ \underbrace{\{s_1, s_2\}}_{\text{Sound}}, \underbrace{\{s_3, s_4\}}_{\text{Distressed}} \right\}, \\
 C_b &= \left\{ \underbrace{\{s_1, s_3\}}_{\text{Compliant}}, \underbrace{\{s_2, s_4\}}_{\text{Non-compl.}} \right\}.
 \end{aligned} \tag{9}$$

The set $\{s_1, s_2\}$ (“Sound”) is not the same as $\{s_1, s_3\}$ (“Compliant”)—no atom appears in both contexts. This

means there is no organizational property that can be ascertained regardless of audit type; *every* piece of information the auditor obtains is context-dependent in the sense that it tells the auditor about one dimension while destroying information about the other. This is *complete complementarity*—stronger than the L_{12} case, where at least one type (the Star) was identifiable under either instrument.

D. Connection to bounded rationality

This formalization connects directly to Herbert Simon’s concept of *bounded rationality* [38]: an external regulator (or market analyst, or journalist) attempting to understand an organization faces an irreducible limitation. The organization’s internal state is fully determinate, but no single probe can reveal it completely, and probing inevitably disturbs the system.

The partition logic captures this as a structural feature, not merely a practical inconvenience. The non-Boolean pasting of the two two-element Boolean algebras means that the auditor’s total knowledge *cannot be represented as a single probability distribution over a single Boolean event space*. This is a formal expression of organizational opacity—the well-known difficulty of assessing complex organizations from the outside [34].

The automaton aspect adds a further layer: issuing directive a causes the organization to *transition to a new state*, meaning that a subsequent probe with b reveals information about the *post-audit* state, not the original one. This is precisely the measurement-disturbance problem of quantum mechanics, realized here by a completely classical mechanism: the audit process itself consumes resources and alters the system.

IX. DISCUSSION: DUAL PROBABILITIES AND QUANTUM COGNITION

A. Summary of the dual interpretation

All six scenarios share a common formal skeleton: a partition logic pasted from Boolean subalgebras over a finite type set S_n . On each such logic, two or more probability theories may coexist:

1. **Classical probabilities** from the convex hull of dispersion-free weights ($VP(G)$). These are the “natural” probabilities for the social science scenarios as described: every applicant, respondent, patient, agent, action, or organization *is* a determinate type, and uncertainty is purely epistemic (arising from the observer’s inability to probe all contexts simultaneously). Classical probabilities obey the Klyachko–Bub–Stairs inequality (6) for the pentagon and analogous inequalities for other graphs.

2. **Quantum-like (Born-rule) probabilities** from a faithful orthogonal representation ($\text{TH}(G)$). These probabilities can exceed the classical bounds—for the pentagon, up to $\sqrt{5}$ instead of 2. They would be appropriate if the “resource inside the black box” were a quantum system rather than a classical partition logic.

B. Why this matters for social science

a. The resource determines the probability theory. The same exclusivity graph (e.g., the pentagon) can be implemented by a Wright urn (yielding $\text{VP}(G)$) or by a quantum system (yielding $\text{TH}(G)$). Similarly, a survey with pentagon structure could yield different probability bounds depending on whether the respondents’ cognitive processes are “classical” (definite beliefs selected by a coarse-grained observation) or “quantum-like” (superposed belief states projected by the survey question). The graph alone does not settle the matter; the *nature of the cognitive resource* does.

b. Complementarity does not imply contextuality. All six scenarios feature complementarity (the inability to perform all measurements simultaneously and consistently; that is, with context-independent valuations [39]), yet every element has a determinate value in every dispersion-free state. The partition logics have a separating set of two-valued states and are therefore *not* contextual in the Kochen–Specker sense [39]. In the social-science analogues, this is intuitive: the applicant has a real competence profile, the patient has a real subtype, the organization is in a real state—the complementarity is epistemic rather than ontic—it is the observer’s problem, not the system’s.

c. Testable predictions. The dual-probability framework suggests concrete experiments. Design a social-measurement scenario with a known exclusivity structure (e.g., five cyclically complementary survey questions). Collect data from many respondents, each answering only one question. Compute the empirical probabilities on the intertwining atoms. If the sum exceeds the classical bound (2 for the pentagon), the classical partition-logic model is falsified, and a quantum-like model is supported. This program extends the work of Busemeyer and Bruza [5] from pairwise order effects to the richer graph-theoretic structures of partition logics.

X. CONCLUSION

We have demonstrated that partition logics—non-Boolean structures originally developed in the founda-

tions of quantum mechanics, generalized urn models, and automata theory [4]—arise naturally in six distinct social-science domains. In each case, a finite set of underlying types (applicant profiles, respondent attitudes, diagnostic subtypes, agent coordination types, legal actions, organizational states) is partitioned by multiple incompatible observational procedures (assessment instruments, survey frames, diagnostic tests, interrogation modes, legal codes, audit directives). The resulting structures exhibit complementarity: no single procedure reveals all relevant distinctions. Yet every system has a fully determinate type; the limitation is epistemic, not ontic.

The social-science scenarios therefore occupy a kind of “purgatory” between classical Boolean and quantum realms: they have non-Boolean logic and complementarity, yet retain full value definiteness [9]. This demonstrates concretely that *complementarity does not imply contextuality*—a point often obscured in discussions that focus exclusively on quantum mechanics.

At the same time, the mathematical fact that many partition logics admit a dual, quantum-like probability interpretation (via faithful orthogonal representations and the Born rule) opens a door to empirical testing. If human cognitive or social systems turn out to produce probability distributions that violate the classical bounds but respect, say, the quantum-like or more exotic ones, the partition-logic framework provides the precise mathematical language in which to formulate and test this hypothesis. Conversely, if the classical bounds are respected, the partition logic with its convex hull of dispersion-free weights provides a complete and parsimonious account of the observed complementarity—no quantum formalism required.

The development of experiments that probe these questions—for instance, large-scale surveys with carefully designed pentagon structures, or clinical assessment batteries with triangle-logic configurations—is a promising direction for future work at the intersection of formal logic, quantum foundations, and social science methodology.

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